

Between Sector Misallocation

Pedro Armada

Halle Institute for Economic Research (IWH)
Friedrich-Schiller University Jena (FSU)

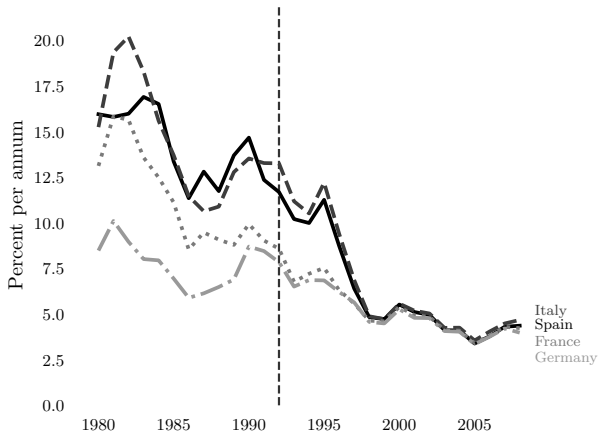
June 26, 2026

Introduction

- The allocation of economic activity across sectors is central to aggregate productivity dynamics.
- Sectoral composition often changes during periods of structural transformation, trade liberalization, and credit booms.
- Whether these reallocations raise or lower productivity depends on the structural differences and frictions that shape how resources move across sectors.

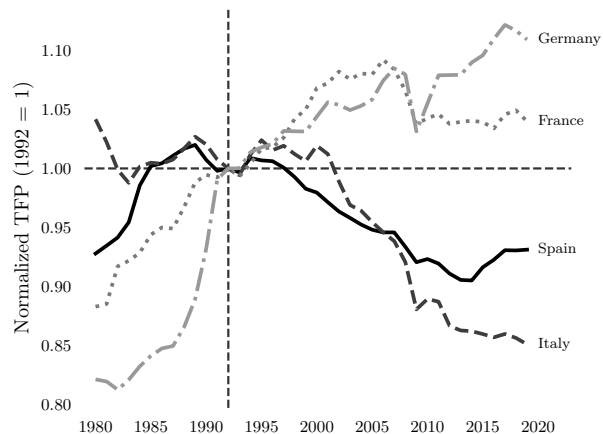
Falling interest rates and declining productivity

Interest Rates on Government Bonds



Source: IMF International Financial Statistics

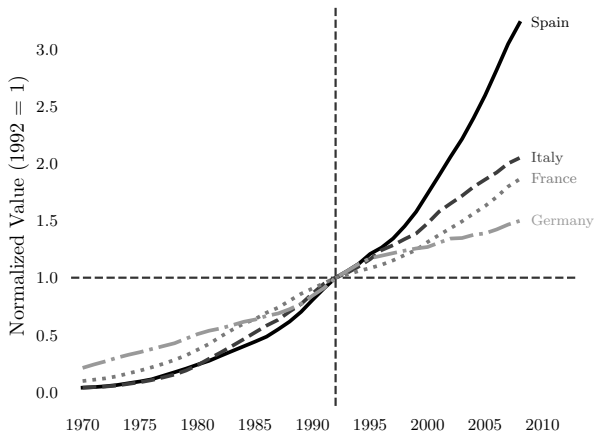
Aggregate Productivity



Source: Penn World Table

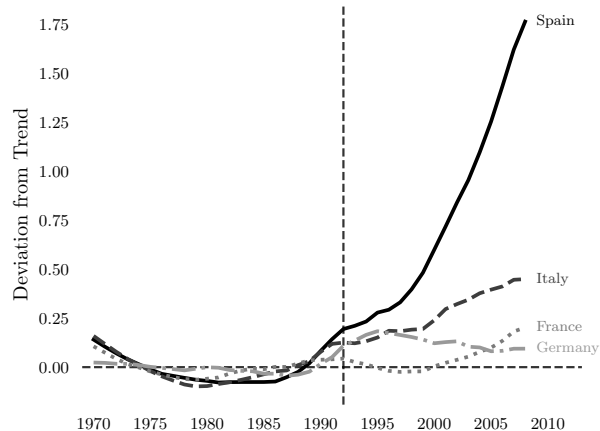
Nontradables expand during the capital inflow episode

Nontradable Output



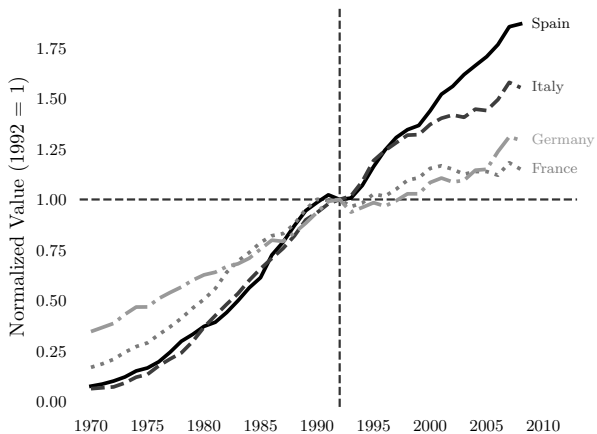
Source: Historical EU KLEMS

Deviation from Trend



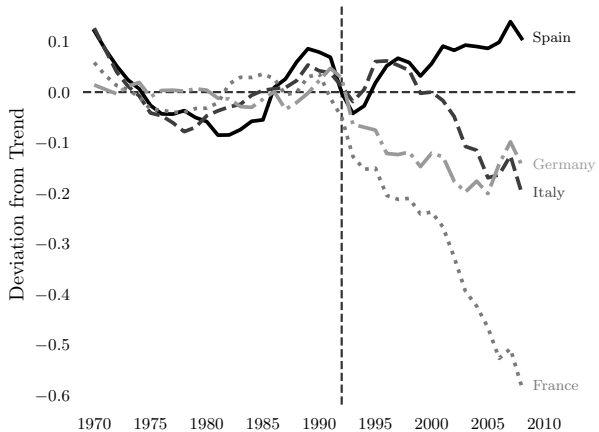
Tradables do not expand commensurately

Tradable Output



Source: Historical EU KLEMS

Deviation from Trend



This Paper

How do sectoral differences determine the allocation and productivity effects of capital inflows?

- Competing explanations:
 - Low competition in nontradables distorts allocation
(Brunnermeier & Reis, 2023; Dias et al., 2016, 2020; Reis, 2013)
 - Capital inflows crowd out high-productivity tradable investment
(Benigno & Fornaro, 2014; Benigno et al., 2025)
 - Tighter borrowing limits in nontradables
(Ozhan, 2020; Schneider & Tornell, 2004; Tornell & Westermann, 2002)
- Understanding these mechanisms is crucial for **policy**

This Paper

- Develop a **multi-sector quantitative model** with rich firm- and sector-level heterogeneity:
 - Sectoral differences in technology, competition, risk
 - Borrowing constraints, capital adjustment costs, sectoral mobility costs
- Let the data discipline the importance of each mechanism by comparing model-simulated and observed dynamics
- I find that **capital frictions** are key to explaining sectoral reallocation and declining allocative efficiency, while competition and credit frictions play a more modest role

Conceptual Framework

- Builds on Hsieh–Klenow: heterogeneous firms produce differentiated varieties under monopolistic competition.
- This paper adds **sectoral heterogeneity** and **sector-wide distortions** to study misallocation across sectors.

Across sectors

$$Y_t = \left(\sum_{s=1}^S Y_{st}^{\frac{\eta-1}{\eta}} \right)^{\frac{\eta}{\eta-1}}$$

Within sectors

$$Y_{st} = \left(\sum_{i=1}^{N_{st}} y_{ist}^{\frac{\theta_s-1}{\theta_s}} \right)^{\frac{\theta_s}{\theta_s-1}}$$

- Firm-level demand:

$$y_{ist} = \left(\frac{p_{ist}}{P_{st}} \right)^{-\theta_s} \left(\frac{P_{st}}{P_t} \right)^{-\eta} Y_t$$

Firm and Sector Wedges

- **Sector-wide wedges** create systematic differences in marginal products **across** sectors, while **firm-level wedges** create dispersion **within** sectors.
- Firm i in sector s chooses price, capital, and labor:

$$\max_{p,k,l} (1 - \tilde{\tau}_{ist}^y) p_{ist} y_{ist} - (1 + \tilde{\tau}_{ist}^k)(r_t + \delta_s) k_{ist} - w_{st} l_{ist}$$

- Composite wedges combine firm-specific and sector-wide distortions:

$$1 - \tilde{\tau}_{ist}^y = (1 - \tau_{ist}^y)(1 - \tau_{st}^y) \quad 1 + \tilde{\tau}_{ist}^k = (1 + \tau_{ist}^k)(1 + \tau_{st}^k)$$

Aggregation and Productivity Losses

- **Aggregate productivity** falls when distortions lower sectoral TFP or shift activity toward sectors with lower productivity.
- Sectoral TFP depends on physical productivity and distorted firm scale:

$$TFP_{st} = \left(\sum_i \left(z_{ist} \frac{TFPR_{st}}{TFPR_{ist}} \right)^{\theta_s - 1} \right)^{\frac{1}{\theta_s - 1}}$$

- Aggregate TFP combines sectoral TFP and sectoral composition:

$$TFP_t = \left(\sum_{s=1}^S \omega_{st} TFP_{st}^{\frac{\eta-1}{\eta}} \right)^{\frac{\eta}{\eta-1}}$$

Illustrative Model

- Simple model to build intuition of how wedges affect both **input choice** and **sectoral sorting**
- Static economy with heterogeneous firms and sectors:

$$y = zk \qquad y = p^{-\theta_s} \qquad \mu_s = \frac{\theta_s}{\theta_s - 1}$$

- Firms first choose a sector, then choose capital before the realized capital wedge is observed:

$$s^* = \arg \max_s \mathbb{E}[\pi(s)]$$

- Conditional on sector s , capital solves:

$$\pi(s) = \max_k \mathbb{E} \left[(zk)^{\frac{\theta_s-1}{\theta_s}} - (1 + \tilde{\tau}^k)rk \right]$$

Effect on Input Choice

- Capital wedges reduce capital relative to the efficient benchmark:

$$\frac{k_s^*}{k_s^e} = \left(\frac{1}{\mathbb{E}(1 + \tilde{\tau}_s^k)} \right)^{\theta_s}$$

- The loss is larger in high-elasticity sectors:

$$\frac{\partial(k_s^*/k_s^e)}{\partial\theta_s} < 0 \quad \text{when} \quad \mathbb{E}(1 + \tilde{\tau}_s^k) > 1$$

- Since

$$\mathbb{E}(1 + \tilde{\tau}_s^k) = 1 + \mathbb{E}(\tau_i^k) + \mathbb{E}(\tau_s^k) + \mathbb{E}(\tau_i^k)\mathbb{E}(\tau_s^k) + \text{Cov}(\tau_i^k, \tau_s^k)$$

positive covariance between firm- and sector-level wedges *amplifies* the capital shortfall.

Effect on Firm Sorting

- Consider two sectors, A and B , with $\theta_A > \theta_B > 1$. A firm chooses the high-elasticity sector A if $z > \bar{z}$.

$$\bar{z} = \underbrace{r}_{\text{cost of capital}} \cdot \underbrace{\left(\frac{\theta_B - 1}{\theta_A - 1} \cdot \frac{\mu_A^{\theta_A}}{\mu_B^{\theta_B}} \right)^{\frac{1}{\theta_A - \theta_B}}}_{\text{scale gains vs. markups}} \cdot \underbrace{\frac{[\mathbb{E}(1 + \tilde{\tau}_A^k)]^{\frac{\theta_A - 1}{\theta_A - \theta_B}}}{[\mathbb{E}(1 + \tilde{\tau}_B^k)]^{\frac{\theta_B - 1}{\theta_A - \theta_B}}}}_{\text{relative capital distortions}}$$

- High-productivity firms sort into high-elasticity sectors because profits scale as $z^{\theta_s - 1}$.
- A higher capital wedge in sector A raises $\bar{z}$ and pushes marginal firms away from the high-elasticity sector.
- Sorting is distorted when the sector that offers the largest scale gains is also the sector with tighter capital frictions.

Effect of Capital Flows

- Model capital inflows as a decline in the cost of capital: $r \downarrow$
- **Intensive margin:** capital expands more in high-elasticity sectors:

$$\frac{\partial \ln k_s^*}{\partial \ln r} = -\theta_s$$

- **Extensive margin:** lower r reduces the productivity threshold for the high-elasticity sector:

$$\bar{z} \propto r \quad \implies \quad \frac{\partial \ln \bar{z}}{\partial \ln r} = 1$$

- Capital inflows therefore operate through two allocation margins:
 - within-sector capital deepening
 - between-sector re-sorting of marginal firms
- Relative sectoral wedges determine whether the reallocation improves or worsens allocative efficiency.

Quantitative Model

- **Objective:** use the quantitative model to evaluate competing channels behind the observed dynamics and quantify their relative importance.
- The model combines heterogeneous firms with sectoral differences in technology, competition, risk, and financial frictions.
- Input wedges arise endogenously from firms' dynamic responses to the constraints they face.
- Capital inflows modeled as exogenous decline in the real interest rate.

Recursive Formulation

- Entrepreneurs differ in sector, net worth, capital, and productivity:

$$\mathbf{x} = (s, a, k) \quad \mathbf{z} = (z^P, z_T^T, z_N^T)$$

- Given aggregate prices $\Gamma = (w, r)$, the entrepreneur solves:

$$V(\mathbf{x}, \mathbf{z}; \Gamma) = \max_{s', c, a', k', l, p} \{u(c) + \beta \mathbb{E}_{\mathbf{z}'} V(\mathbf{x}', \mathbf{z}'; \Gamma')\}$$

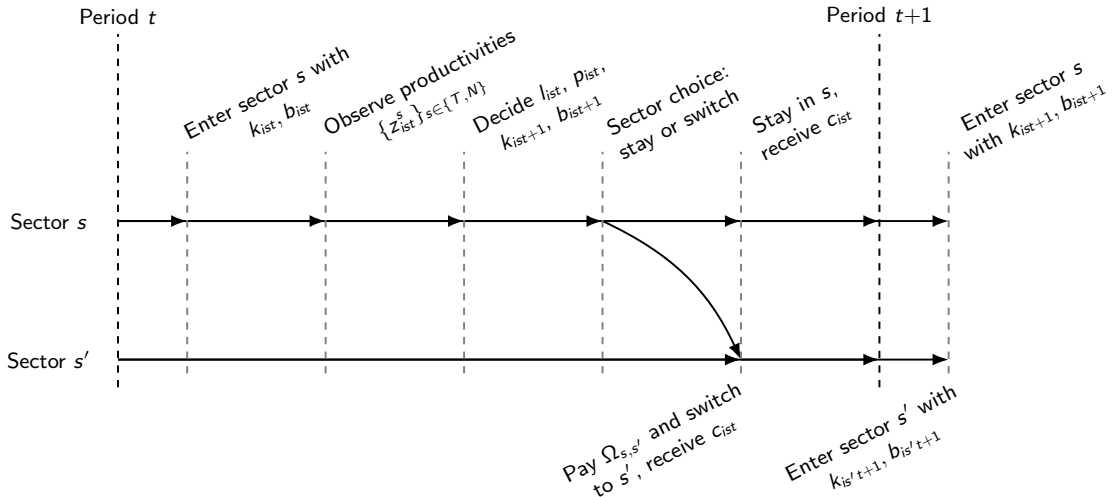
$$c + a' + \underbrace{\Phi_s(k', k)}_{\text{capital adjustment cost}} + \underbrace{\Omega_{s,s'}(k)}_{\text{sector mobility cost}} = p_s y_s - wl - (r + \delta)k - \kappa_s + (1 + r)a$$

$$y_s = z_s k^{\alpha_s} l^{1-\alpha_s}$$

$$y_s = p_s^{-\theta_s}$$

$$k' \leq \lambda_s a'$$

Timing Protocol



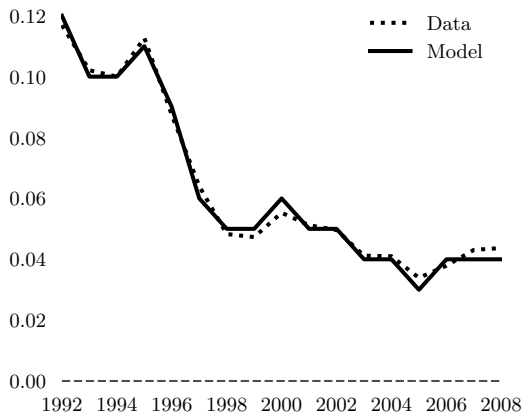
Transition Experiment

- **Objective:** calibrate the model to match Spain's post-Maastricht transition.
- For each candidate parameter vector Θ :
 1. solve the stationary equilibrium under the 1992 real interest rate;
 2. simulate the transition using the observed path of real interest rates;
 3. compare model-generated paths with empirical paths.
- Target moments are log changes in TFP_t , Y_t^T , Y_t^N
- The calibrated parameters minimize the distance between model and data:

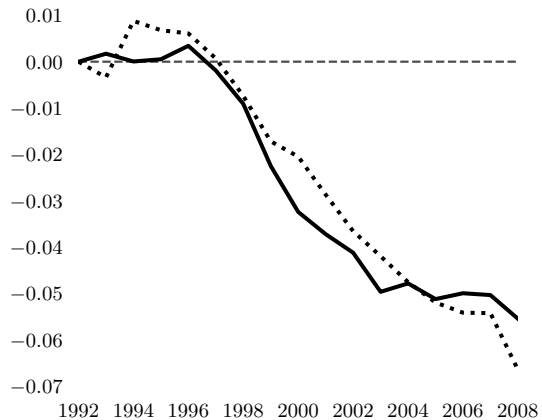
$$\Theta^* = \arg \min_{\Theta} M(\Theta)'WM(\Theta)$$

Quantitative Fit

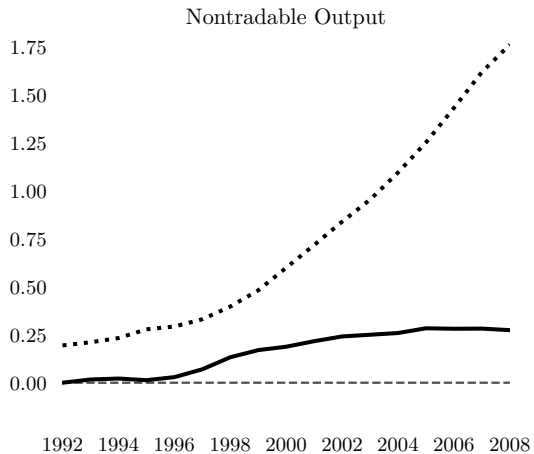
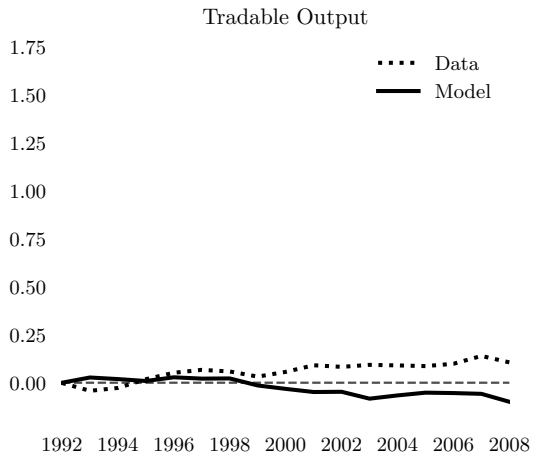
Interest Rate



TFP



Quantitative Fit



Quantitative Fit

- Modest sectoral differences in **competition** and **credit constraints**:
 - Demand elasticity (θ) 10% higher in tradables
 - Borrowing constraint (λ) 9% more relaxed in tradables
- Larger asymmetries in **capital frictions**:
 - Capital adjustment costs (ϕ) 53% higher in tradables
 - Sector switching costs (ω) 55% higher in tradables
- Capital frictions dampen tradable sector expansion in response to falling r :
 - Capital adjustment costs limit *intensive* margin
 - Mobility costs limit *extensive* margin

Conclusion

- Capital inflows can reduce productivity by shifting activity toward nontradables, but the underlying mechanisms remain unclear
- I develop a multi-sector model with rich heterogeneity to quantify sources of sectoral reallocation and TFP decline
- I find that **capital frictions** play a central role in explaining the observed dynamics
- From a **policy** perspective, facilitating capital mobility can raise productivity by improving reallocation of inputs

Extra Slides

CES Aggregation

Final good aggregation

$$Y_t = \left(\sum_{s=1}^S Y_{st}^{\frac{\eta-1}{\eta}} \right)^{\frac{\eta}{\eta-1}} \quad \eta > 1$$

Aggregate price index

$$P_t = \left(\sum_{s=1}^S P_{st}^{1-\eta} \right)^{\frac{1}{1-\eta}}$$

Sectoral expenditure shares

$$\frac{P_{st} Y_{st}}{P_t Y_t} = \left(\frac{P_{st}}{P_t} \right)^{1-\eta}$$

Why CES Across Sectors?

Cobb–Douglas across sectors

$$Y_t = \prod_s Y_{st}^{\eta_s} \quad P_{st} Y_{st} = \eta_s P_t Y_t$$

- Expenditure shares fixed.
- Sectoral productivity changes absorbed by prices.
- Input allocation across sectors does not respond through expenditure shares.

CES across sectors

$$Y_t = \left(\sum_s Y_{st}^{\frac{\eta-1}{\eta}} \right)^{\frac{\eta}{\eta-1}} \quad P_t = \left(\sum_s P_{st}^{1-\eta} \right)^{\frac{1}{1-\eta}}$$

- Expenditure shares respond to relative prices.
- Sectoral productivity affects sectoral size.
- Departure necessary to study intersectoral allocation.

Firm Demand

Within-sector aggregation

$$Y_{st} = \left(\sum_{i=1}^{N_{st}} y_{ist}^{\frac{\theta_s-1}{\theta_s}} \right)^{\frac{\theta_s}{\theta_s-1}} \quad \theta_s > 1$$

Sectoral price index

$$P_{st} = \left(\sum_{i=1}^{N_{st}} p_{ist}^{1-\theta_s} \right)^{\frac{1}{1-\theta_s}}$$

Firm-level demand

$$y_{ist} = \left(\frac{p_{ist}}{P_{st}} \right)^{-\theta_s} \left(\frac{P_{st}}{P_t} \right)^{-\eta} Y_t$$

Firm First-Order Conditions

Firm problem

$$\max_{p_{ist}, k_{ist}, l_{ist}} (1 - \tilde{\tau}_{ist}^y) p_{ist} y_{ist} - (1 + \tilde{\tau}_{ist}^k)(r_t + \delta_s) k_{ist} - w_{st} l_{ist}$$

Production

$$y_{ist} = z_{ist} k_{ist}^{\alpha_s} l_{ist}^{1-\alpha_s}$$

FOC

$$MRPL_{ist} = \left(\frac{1 - \alpha_s}{\mu_s} \right) \frac{p_{ist} y_{ist}}{l_{ist}} = \frac{1}{1 - \tilde{\tau}_{ist}^y} w_{st}$$

$$MRPK_{ist} = \left(\frac{\alpha_s}{\mu_s} \right) \frac{p_{ist} y_{ist}}{k_{ist}} = \frac{1 + \tilde{\tau}_{ist}^k}{1 - \tilde{\tau}_{ist}^y} (r_t + \delta_s)$$

TFPR and Marginal Products

Revenue productivity

$$TFPR_{ist} = p_{ist} z_{ist} = \frac{p_{ist} y_{ist}}{k_{ist}^{\alpha_s} l_{ist}^{1-\alpha_s}}$$

Equivalent expression using marginal products

$$TFPR_{ist} = \mu_s \left(\frac{MRPK_{ist}}{\alpha_s} \right)^{\alpha_s} \left(\frac{MRPL_{ist}}{1 - \alpha_s} \right)^{1-\alpha_s}$$

Substituting the FOCs

$$TFPR_{ist} = \mu_s \left[\frac{1 + \tilde{\tau}_{ist}^k}{1 - \tilde{\tau}_{ist}^y} \frac{r_t + \delta_s}{\alpha_s} \right]^{\alpha_s} \left[\frac{1}{1 - \tilde{\tau}_{ist}^y} \frac{w_{st}}{1 - \alpha_s} \right]^{1-\alpha_s}$$

Sectoral TFP

Sectoral inputs

$$K_{st} = \sum_{i=1}^{N_{st}} k_{ist} \quad L_{st} = \sum_{i=1}^{N_{st}} l_{ist}$$

Sectoral TFP

$$TFP_{st} = \frac{Y_{st}}{K_{st}^{\alpha_s} L_{st}^{1-\alpha_s}} = \left(\sum_{i=1}^{N_{st}} \left(z_{ist} \frac{TFPR_{st}}{TFPR_{ist}} \right)^{\theta_s - 1} \right)^{\frac{1}{\theta_s - 1}}$$

Efficient benchmark

$$TFPR_{ist} = TFPR_{st} \quad \forall i \quad \Rightarrow \quad TFP_{st}^e = \left(\sum_{i=1}^{N_{st}} z_{ist}^{\theta_s - 1} \right)^{\frac{1}{\theta_s - 1}}$$

TFP Loss from Misallocation

Distorted sectoral TFP

$$TFP_{st} = \left(\sum_i \left(z_{ist} \frac{TFPR_{st}}{TFPR_{ist}} \right)^{\theta_s - 1} \right)^{\frac{1}{\theta_s - 1}}$$

Efficient sectoral TFP

$$TFP_{st}^e = \left(\sum_i z_{ist}^{\theta_s - 1} \right)^{\frac{1}{\theta_s - 1}}$$

Log TFP loss

$$\log \left(\frac{TFP_{st}}{TFP_{st}^e} \right) = \frac{1}{\theta_s - 1} \left[\log \left(\sum_i z_{ist}^{\theta_s - 1} \left(\frac{TFPR_{st}}{TFPR_{ist}} \right)^{\theta_s - 1} \right) - \log \left(\sum_i z_{ist}^{\theta_s - 1} \right) \right]$$

Aggregate TFP

Aggregate inputs

$$K_t = \sum_{s=1}^S K_{st} \quad L_t = \sum_{s=1}^S L_{st}$$

Aggregate output

$$Y_t = \left(\sum_{s=1}^S Y_{st}^{\frac{\eta-1}{\eta}} \right)^{\frac{\eta}{\eta-1}} = \left(\sum_{s=1}^S \left[TFP_{st} K_{st}^{\alpha_s} L_{st}^{1-\alpha_s} \right]^{\frac{\eta-1}{\eta}} \right)^{\frac{\eta}{\eta-1}}$$

Aggregate TFP

$$TFP_t = \left(\sum_{s=1}^S \omega_{st} TFP_{st}^{\frac{\eta-1}{\eta}} \right)^{\frac{\eta}{\eta-1}} \quad \omega_{st} = \frac{Y_{st}^{\frac{\eta-1}{\eta}}}{\sum_{s'=1}^S Y_{s't}^{\frac{\eta-1}{\eta}}}$$

Capital Wedges

Composite capital wedge

$$1 + \tilde{\tau}_s^k = (1 + \tau_i^k)(1 + \tau_s^k)$$

Expected capital wedge

$$\begin{aligned}\mathbb{E}(1 + \tilde{\tau}_s^k) &= \mathbb{E} \left[(1 + \tau_i^k)(1 + \tau_s^k) \right] \\ &= 1 + \mathbb{E}(\tau_i^k) + \mathbb{E}(\tau_s^k) + \mathbb{E}(\tau_i^k \tau_s^k)\end{aligned}$$

Using covariance

$$\mathbb{E}(1 + \tilde{\tau}_s^k) = 1 + \mathbb{E}(\tau_i^k) + \mathbb{E}(\tau_s^k) + \mathbb{E}(\tau_i^k)\mathbb{E}(\tau_s^k) + \text{Cov}(\tau_i^k, \tau_s^k)$$

Correlation Between Wedges

Composite capital wedge

$$1 + \tilde{\tau}_s^k = (1 + \tau_i^k)(1 + \tau_s^k)$$

Expected wedge

$$\mathbb{E}(1 + \tilde{\tau}_s^k) = 1 + \mathbb{E}(\tau_i^k) + \mathbb{E}(\tau_s^k) + \mathbb{E}(\tau_i^k)\mathbb{E}(\tau_s^k) + \text{Cov}(\tau_i^k, \tau_s^k)$$

Perfect positive correlation

$$\tau_s^k = \tau_i^k \quad \Rightarrow \quad \mathbb{E}(1 + \tilde{\tau}_s^k) = 1 + 2\mathbb{E}(\tau_i^k) + \mathbb{E}(\tau_i^k)^2 + \text{Var}(\tau_i^k)$$

Perfect negative correlation

$$\tau_s^k = -\tau_i^k \quad \Rightarrow \quad \mathbb{E}(1 + \tilde{\tau}_s^k) = 1 - \mathbb{E}[(\tau_i^k)^2]$$

Optimal Capital Choice

Firm problem

$$\max_k \mathbb{E} \left[(zk)^{\frac{\theta_s-1}{\theta_s}} - (1 + \tilde{\tau}_s^k)rk \right]$$

FOC

$$\frac{\theta_s - 1}{\theta_s} z^{\frac{\theta_s-1}{\theta_s}} k^{-\frac{1}{\theta_s}} = r\mathbb{E}(1 + \tilde{\tau}_s^k)$$

Optimal capital

$$k_s^* = \left[\frac{\frac{\theta_s-1}{\theta_s} z^{\frac{\theta_s-1}{\theta_s}}}{r\mathbb{E}(1 + \tilde{\tau}_s^k)} \right]^{\theta_s} = \frac{z^{\theta_s-1}}{\mu_s^{\theta_s} r^{\theta_s} [\mathbb{E}(1 + \tilde{\tau}_s^k)]^{\theta_s}}$$

with $\mu_s = \theta_s/(\theta_s - 1)$

Capital Gap

Efficient capital choice

$$k_s^e = \frac{z^{\theta_s-1}}{\mu_s^{\theta_s} r^{\theta_s}}$$

Capital relative to efficient benchmark

$$\frac{k_s^*}{k_s^e} = \left(\frac{1}{\mathbb{E}(1 + \tilde{\tau}_s^k)} \right)^{\theta_s} = \left[\frac{1}{1 + \mathbb{E}(\tau_i^k) + \mathbb{E}(\tau_s^k) + \mathbb{E}(\tau_i^k)\mathbb{E}(\tau_s^k) + \text{Cov}(\tau_i^k, \tau_s^k)} \right]^{\theta_s}$$

Effect of sectoral demand elasticity

$$\frac{\partial(k_s^*/k_s^e)}{\partial\theta_s} = - \left[\mathbb{E}(1 + \tilde{\tau}_s^k) \right]^{-\theta_s} \log \left[\mathbb{E}(1 + \tilde{\tau}_s^k) \right] < 0$$

Optimal Profits

Profit at optimal capital

$$\pi^*(s) = \max_k \mathbb{E} \left[(zk)^{\frac{\theta_s-1}{\theta_s}} - (1 + \tilde{r}_s^k)rk \right]$$

Using the optimal capital choice

$$k_s^* = \frac{z^{\theta_s-1}}{\mu_s^{\theta_s} r^{\theta_s} [\mathbb{E}(1 + \tilde{r}_s^k)]^{\theta_s}}$$

Expected profit

$$\pi^*(s) = \frac{(\theta_s - 1)^{\theta_s-1} z^{\theta_s-1}}{\theta_s^{\theta_s} r^{\theta_s-1} [\mathbb{E}(1 + \tilde{r}_s^k)]^{\theta_s-1}}$$

Productivity elasticity of profits

$$\frac{\partial \ln \pi^*(s)}{\partial \ln z} = \theta_s - 1$$

Sorting Threshold

Sector comparison

$$\theta_A > \theta_B > 1 \quad \text{and} \quad \pi^*(A) > \pi^*(B)$$

Expected profits

$$\pi^*(s) = \frac{(\theta_s - 1)^{\theta_s - 1} z^{\theta_s - 1}}{\theta_s^{\theta_s} r^{\theta_s - 1} [\mathbb{E}(1 + \tilde{\tau}_s^k)]^{\theta_s - 1}}$$

Threshold productivity

$$\bar{z} = r \left(\frac{\theta_B - 1}{\theta_A - 1} \frac{\mu_A^{\theta_A}}{\mu_B^{\theta_B}} \right)^{\frac{1}{\theta_A - \theta_B}} \frac{[\mathbb{E}(1 + \tilde{\tau}_A^k)]^{\frac{\theta_A - 1}{\theta_A - \theta_B}}}{[\mathbb{E}(1 + \tilde{\tau}_B^k)]^{\frac{\theta_B - 1}{\theta_A - \theta_B}}}$$

Sorting rule

$$z > \bar{z} \quad \Rightarrow \quad s^* = A$$

Capital Inflows

Capital inflow shock: $r \downarrow$

Intensive margin

$$k_s^* = \frac{z^{\theta_s - 1}}{\mu_s^{\theta_s} r^{\theta_s} [\mathbb{E}(1 + \tilde{\tau}_s^k)]^{\theta_s}} \Rightarrow \frac{\partial \ln k_s^*}{\partial \ln r} = -\theta_s$$

Extensive margin

$$\bar{z} \propto r \Rightarrow \frac{\partial \ln \bar{z}}{\partial \ln r} = 1$$

$r \downarrow \Rightarrow$ capital expands within sectors and the sorting threshold falls.

Model Primitives

Environment

- Small open economy populated by a large number of monopolistically competitive firms
- Firms hire labor competitively at wage w_t
- Firms save and borrow internationally through a one-period bond traded at interest rate r_t
- No aggregate uncertainty

Entrepreneur preferences

$$\mathbb{E} \sum_{t=0}^{\infty} \beta^t \left(\frac{c_{ist}^{1-\gamma} - 1}{1-\gamma} \right)$$

Technology and Demand

Production

$$y_s = z_s k^{\alpha_s} l^{1-\alpha_s}$$

Demand

$$y_s = p_s^{-\theta_s} \quad \Longleftrightarrow \quad p_s = y_s^{-\frac{1}{\theta_s}}$$

Markup

$$\mu_s = \frac{\theta_s}{\theta_s - 1}$$

Operating profits

$$\pi_s = p_s y_s - w l - \kappa_s$$

Productivity Process

Firm productivity

$$z_{ist} = z_i^P \exp(z_{ist}^T)$$

Permanent productivity

$$F(z_i^P) = 1 - \left(\frac{1}{z_i^P} \right)^\varphi \quad z_i^P \geq 1$$

Sector-specific transitory productivity

$$z_{ist}^T = \rho_s z_{ist-1}^T + \varepsilon_{ist}^T \quad \varepsilon_{ist}^T \sim N(0, \sigma_s^T)$$

Sectoral Frictions

Borrowing constraint

$$b_{ist} \leq \chi_s k_{ist} \quad \Longleftrightarrow \quad k' \leq \lambda_s a' \quad \lambda_s = \frac{1}{1 - \chi_s}$$

Capital adjustment cost

$$\Phi_s(k', k) = \frac{\phi_s}{2} \left(\frac{k' - (1 - \delta)k}{k} \right)^2 k$$

Sectoral mobility cost

$$\Omega_{s,s'}(k) = \mathbb{I}_{s' \neq s} \omega_s k$$

Optimal Labor Demand

Static operating profits

$$\pi_s = \left(z_s k^{\alpha_s} l^{1-\alpha_s} \right)^{\frac{\theta_s-1}{\theta_s}} - w l - \kappa_s$$

Labor FOC

$$\frac{\theta_s - 1}{\theta_s} (1 - \alpha_s) z_s^{\frac{\theta_s-1}{\theta_s}} k^{\frac{\alpha_s(\theta_s-1)}{\theta_s}} l^{-\frac{1+\alpha_s(\theta_s-1)}{\theta_s}} = w$$

Optimal labor demand

$$l_s(k, z_s, w) = \left(\frac{(1 - \alpha_s)(\theta_s - 1)}{w \theta_s} \right)^{\frac{\theta_s}{1+\alpha_s(\theta_s-1)}} z_s^{\frac{\theta_s-1}{1+\alpha_s(\theta_s-1)}} k^{\frac{\alpha_s(\theta_s-1)}{1+\alpha_s(\theta_s-1)}}$$

Budget Constraint

Original budget constraint

$$c + k' - (1 - \delta)k + (1 + r)b + \Phi_s(k', k) + \Omega_{s,s'}(k) = \pi_s + b'$$

Net worth

$$a = k - b \quad a' = k' - b'$$

Recursive form

$$c + a' + \Phi_s(k', k) + \Omega_{s,s'}(k) = \pi_s - (r + \delta)k + (1 + r)a$$

Using operating profits

$$c + a' + \Phi_s(k', k) + \Omega_{s,s'}(k) = p_s y_s - wl - (r + \delta)k - \kappa_s + (1 + r)a$$

Stationary Equilibrium

Given prices $\Gamma = (w, r)$, a stationary partial equilibrium satisfies:

1. **Optimality:** The value function solves:

$$V(\mathbf{x}, \mathbf{z}, \Gamma) = \max_{s', c, a', k', l, p} \{u(c) + \beta \mathbb{E}_{\mathbf{z}'} V(\mathbf{x}', \mathbf{z}', \Gamma')\}$$

2. **Feasibility:** Policies satisfy the budget constraint, borrowing constraint, and non-negative net worth condition
3. **Consistency:** Policies map current endogenous states into next-period states:

$$\mathbf{x}' = (s'(\cdot), a'(\cdot), k'(\cdot))$$

4. **Invariant distribution:** The firm distribution is stationary:

$$\mu'(\mathbf{x}', \mathbf{z}') = \mathcal{T}(\mu(\mathbf{x}, \mathbf{z})) \quad \mu' = \mu$$

Simulated Method of Moments

Candidate parameter vector

$$\Theta = (\theta_s, \alpha_s, \kappa_s, \rho_s, \sigma_s, \lambda_s, \phi_s, \omega_s)_{s \in \{T, N\}}$$

Model and data moments

$$\mathcal{M}_t^j(\Theta) \quad \mathcal{D}_t^j \quad j = 1, \dots, J \quad t = 1, \dots, T$$

Moment errors

$$M(\Theta) = \begin{bmatrix} \mathcal{M}_1^1(\Theta) - \mathcal{D}_1^1 \\ \mathcal{M}_2^1(\Theta) - \mathcal{D}_2^1 \\ \vdots \\ \mathcal{M}_T^J(\Theta) - \mathcal{D}_T^J \end{bmatrix}$$

Objective function

$$\Theta^* = \arg \min_{\Theta} M(\Theta)' W M(\Theta) \quad W = I$$

Model Parameters

Assigned Parameters

β	γ	δ	η	φ
0.875	1.50	0.06	1.50	30.0

Fit by Target

Target	TFP_t	Y_t^T	Y_t^N	Total
RMSE	0.006	0.118	0.717	0.420

Fitted Parameters

Parameter	Tradable	Nontradable	% Diff.
Demand elasticity θ_S	4.15	3.75	10%
Capital share α_S	0.34	0.29	16%
Fixed cost κ_S	0.11	0.00	200%
Persistence ρ_S	0.53	0.62	16%
Volatility σ_S	0.28	0.36	25%
Borrowing limit λ_S	1.12	1.02	9%
Adjustment cost ϕ_S	4.51	2.62	53%
Mobility cost ω_S	0.14	0.08	55%